SAKLOLO: A Hybrid Semantic-Probabilistic Framework for Analysing Social Media Emergency Feeds in Bulacan and CAMANAVA

Undergraduate Research Proposal – Title Defense

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Problem

- Emergency social media feeds are high-volume, unstructured, and time-sensitive.
- Critical rescue, damage, and sentiment signals are buried in noisy Taglish posts.
- Manual monitoring is too slow for DRRMOs in Bulacan and CAMANAVA.

Solution: SAKLOLO

System A – Hybrid baseline

- Count Vectorization
- LDA topics
- XLM-RoBERTa sentiment
- Joint Matrix

System B – LLM comparison

- Qwen3-1.7B-Instruct-Q4_K_M
- Few-shot prompting
- Matched topic count
- Joint Matrix output

Main idea

Benchmark a lightweight interpretable hybrid system against a modern open-weight LLM on the same Taglish disaster corpus.

SOP: Statement of the Problem

1. How does System B perform relative to System A in sentiment classification?
2. How strongly are latent topics associated with sentiment in each system?
3. Is there a significant operational difference between the outputs in blind stakeholder evaluation?

Focus

Agreement, association, and usability – not just accuracy.

Scope

Included

- Public Facebook posts/comments
- Bulacan + CAMANAVA
- 2020–2025
- Tagalog, English, Taglish
- 10,000–15,000+ records
- Offline local dashboard

Excluded

- Real-time DRRM integration
- Private posts
- Cloud deployment
- Rescue dispatch simulation
- Enterprise multi-user system

Significance

- **DRRMOs:** faster triage of emergency feeds.
- **LGUs:** localized sentiment and topic insights.
- **Responders:** organized field information.
- **Researchers:** benchmark for Taglish disaster NLP.

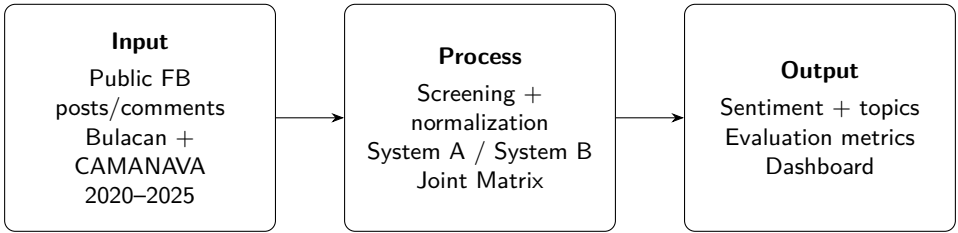
Value

SAKLOLO supports decision-making by converting chaotic social feeds into structured topic–sentiment evidence.

Quick RRL

- **Global:** LDA and transformer hybrids cluster disaster text in floods and typhoons.
- **Philippines:** Social media aids response and recovery, but noise and misinformation remain problems.
- **Local:** Philippine LDA studies exist, but mostly use formal posts, tweets, or non-emergency Taglish feeds.
- **BulSU:** Related theses focus on GPS, alerts, and location sharing, not NLP triage.
- **Gap:** No rigorous benchmark of hybrid joint matrix vs. open-weight LLM for Bulacan/CAMANAVA emergency feeds.

Conceptual Framework



Logic

Same corpus → two systems → same matrix shape → fair comparison.

Research Method

- Comparative quantitative evaluation design.
- Same prepared corpus for System A and System B.
- System A is the relative reference; no absolute ground truth is claimed.

RQ	Metrics	Threshold
RQ1	Precision, Recall, Macro-F1	Macro-F1 ≥ 0.70
RQ1	Cohen's κ	≥ 0.60
RQ2	Chi-square, Cramér's V	$p < 0.05$, $V > 0.07$
RQ3	Agreement rate, McNemar's test	$\geq 70\%$, $\alpha = 0.05$

System Development: Waterfall

1. **Requirements:** fix corpus, metrics, and output format.
2. **Design:** define System A/B pipelines and dashboard.
3. **Implementation:** build collection, preprocessing, models, and UI.
4. **Testing:** run both systems, compute statistics, conduct blind evaluation.
5. **Documentation:** report results, limitations, and handover materials.

Why Waterfall?

The comparison rules are fixed before development to preserve validity.

Respondents

Cohort A: DRRM officers

- 5–10 active practitioners
- Bulacan/CAMANAVA offices
- Judge operational usefulness

Cohort B: technical experts

- 5–10 faculty/researchers
- Math, CS, data analytics
- Judge coherence and validity

Blind panel

Total: 10–20 evaluators; outputs labeled only as X and Y.

Data Gathering Procedure

1. Identify public disaster-related Facebook sources in Bulacan and CAMANAVA.
2. Extract posts/comments using semi-automated scraping and manual screening.
3. Apply inclusion/exclusion criteria.
4. Normalize text, remove duplicates, and anonymize identifiers.
5. Run the same corpus through System A and System B.
6. Generate joint matrices and compute evaluation metrics.
7. Present blinded outputs to the stakeholder panel through the dashboard.

Ethical Consideration

- Only publicly accessible posts/comments are collected.
- Personal identifiers and usernames are removed.
- Data handling follows RA 10173, the Data Privacy Act of 2012.
- Evaluators give voluntary participation and informed consent.
- Results are reported transparently for academic use only.

Thank You

Questions from the Panel

SAKLOLO – Group 10, Bulacan State University